

Solution Architecture

ComicCraft – AI Comic Story Creator

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Architecture overview

The browser renders Jinja2-based pages and submits requests to FastAPI endpoint routers. Backend services validate input, coordinate asynchronous AI generation where supported, assemble panel assets, and produce the final PDF.

Components

Layer	Responsibility
Web UI (Jinja2)	Prompt form, status feedback, preview/download links.
FastAPI routers	Request handling, validation, response and download endpoints.
Generation service	Gemini outline/script calls and Stable Diffusion image generation.
Layout builder	Orders panels and arranges image, narration, and dialogue.
PDF service	Compiles output using FPDF and returns downloadable file.
Configuration	Loads provider credentials from environment; never ships secrets to browser.

Resilience and security

- Apply input validation, request limits, and safe error messages.
- Use provider timeouts and controlled retries; account for rate limits.
- Store temporary generated assets only as long as needed.
- Avoid recording API keys and sensitive prompts in logs.
- The end-to-end generation sequence may exceed a short synchronous web-request window; production deployments should consider background jobs and progress polling.